

A Statistical Study of Amazon Sales, Demand and Revenue

MA5755 Final Project Report

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1 Introduction

In e-commerce (like Amazon), companies give discounts to attract customers, increase sales and compete with other companies. In this project, we investigate the relationship between discounts, sales, revenue and profit in an online Amazon dataset. Instead of assuming directly that discount is good, we analyze sales behavior, how revenue changes with sales and how profit changes across time, category, and region. Does the discount actually increase sales enough to compensate for the lower price? We investigate that **how do discounts affect profit and total revenue for different product categories across regions over time, and do sales increase or decrease when discounts are applied?**

2 Data Description

The dataset used in this study is Amazon Sales Analysis. It is a cleaned datasets. This dataset contains information about the Amazon online product sales. Each row represents a single order made by a customer. It includes details such as the date on which the order has been placed, the product category, price, discount, quantity sold, revenue, and profit. The dataset also records the region of the customer, the payment method, and customer feedback like ratings and review counts.

Dataset Variables are: order_id, order_date, product_id, product_category, price, discount_percent, quantity_sold, customer_region, payment_method, rating, review_count, discounted_price, total_revenue, profit.

- 50,000 orders placed
- Time period: 01-Jan-2022– 31-Dec-2023
- 6 product categories (Beauty, Books, Electronics, Fashion, Home & Kitchen and Sports)

- 4 customer regions (Asia, Europe, Middle East and North America)
- 4000 unique products

3 Preliminary Studies

- **Exploratory Data Analysis:** used for finding the patterns, anomalies and outliers. Here we have checked the distributions of each columns and tried to find out any presence of outliers.
- **Feature Selection:** for carrying out regression analysis, here we have used a few predictors like discount_percent, price, rating, review_count, product_category, customer_region. Because we are trying to find total quantity sold and total revenue.
- **PCA:** used for visualization of the clusters in 2D.

3.1 Exploratory Data Analysis

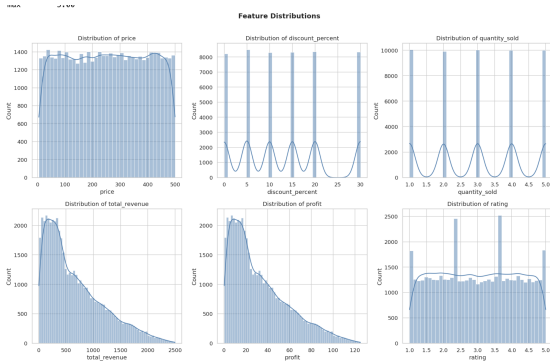


Figure 1: Features distributions

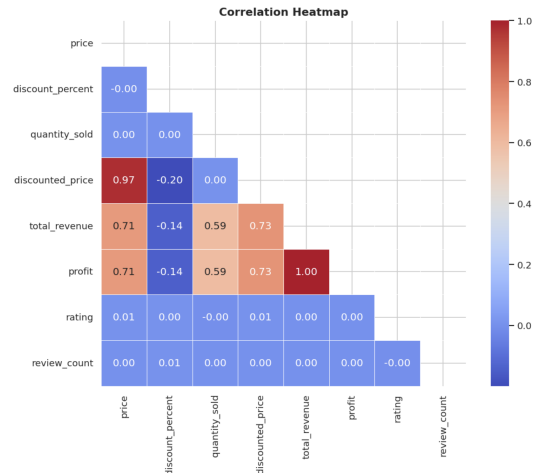


Figure 2: Correlation map

From **figure 1**, we can see the distribution of each feature, and from **figure 2** we can see the correlation between each feature. Total revenue and price are highly correlated to profit and discounted price, respectively. And discount percentage is negatively correlated by profit, total revenue and discounted price.

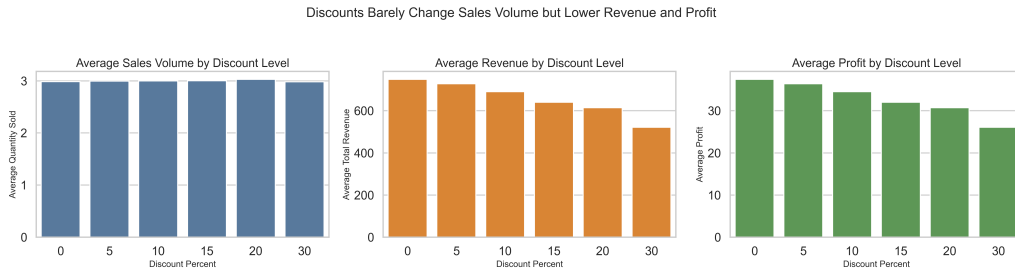


Figure 3: Effect of discounts on sales, revenue, and profit

Here we try to look at what happens on the average as Discount increases. we see from **(figure 3)** that when the discount level increases, the average quantity sold is almost the same for every level of discount. And average revenue decreases as discount increases because the sale is constant.

If sales volume increases strongly with increasing discount, then discounting may be working.

As we can observe from **figure 3**:

- The average quantity sold remains nearly constant (≈ 2.986 to 3.029)
- The average revenue decreases significantly (≈ 749 to 522)
- Correlation (discount vs quantity): 0.0014 , which is almost 0 . It means that there is no relationship between Discount and Number of quantity sold because as increases discount does not increase the sale.
- Correlation (discount vs revenue): -0.1444 , because increasing Discount decreases revenue as sale is constant.
- Correlation (discount vs. profit): -0.1444 , because increasing Discount decreases Profit as the sale is constant.

4 Statistical Analysis

4.1 Discount Analysis

Effect of discount

Discounts clearly reduce the price, but the increase in demand is negligible.

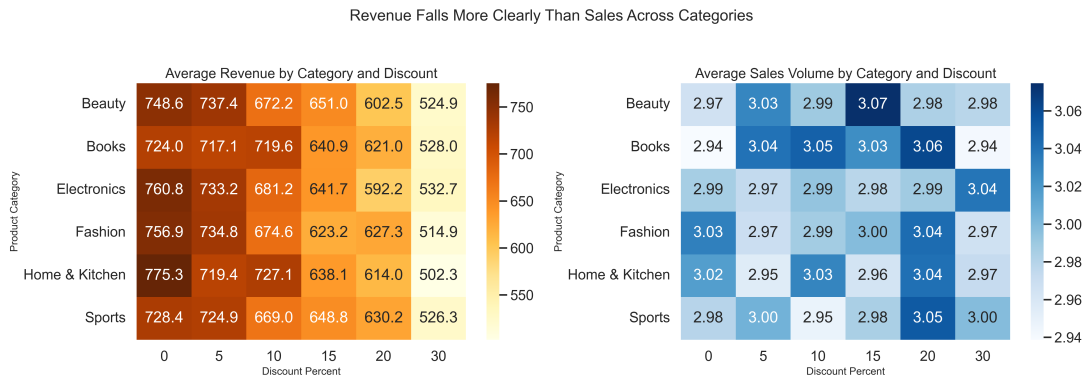


Figure 4: Category-level discount patterns

Also, from the heatmap **(Figure 4)** we can see that revenue decreases across almost all categories as discounts increase. Sales volume remains stable across discount levels within categories.

The weak demand response is not isolated to a single category.

Regional and Category Distribution

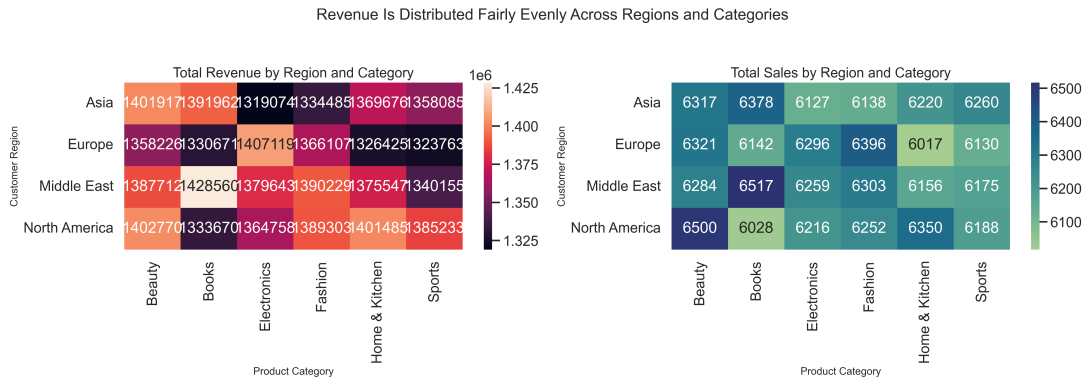


Figure 5: Revenue and sales distribution across regions and categories

From **figure 5**, we can see that revenue and sales are evenly distributed across all four regions. No single region dominates the dataset.

The dataset is balanced and gives conclusions that are more reliable.

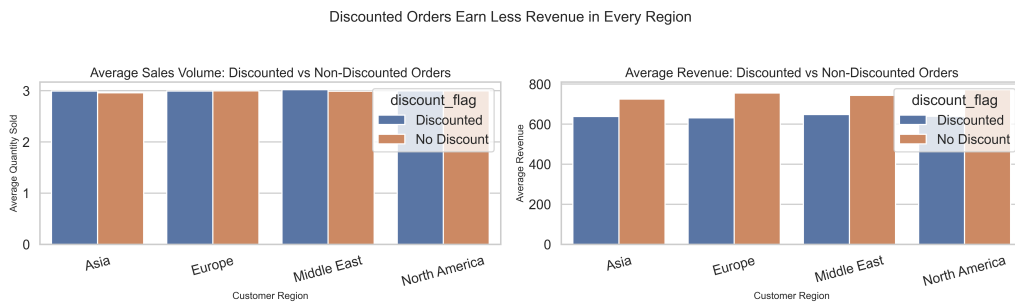


Figure 6: Discounted vs non-discounted comparison across regions

From **figure 6**, we can see that discounted orders do not increase sales volume and discounted orders generate lower revenue in all regions.

The negative impact of discounts is geographically consistent.

4.2 Clustering Analysis

Region-Month Clustering

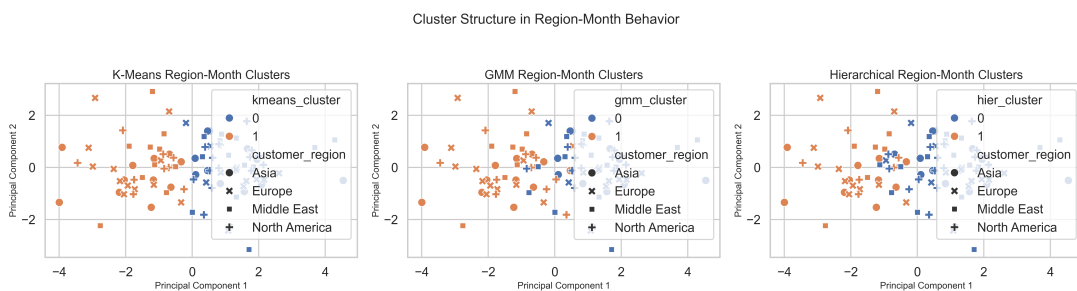


Figure 7: Cluster visualization

Now, from clustering, we try to find some patterns across regions, which helps us to understand which months are strong or weak across regions. We cluster the data month by month to identify groups of region-month observation that share similar revenue, sales, price and discount patterns.

We applied three different clustering methods (K means, GMM, Hierarchical). Cluster 0 has a higher performance (359K revenue, 1628 sales) than Cluster 1 (322K revenue, 1481 sales). From **figure 7**, we can see that all three methods give a similar pattern of clustering. We can see that the Silhouette Scores that separation is clear and stable. From Table 1, the best grouping is at $k = 2$, with the highest silhouette score (K-Means 0.34), indicating a clear grouping.

k	K-Means	GMM	Hierarchical
2	0.340	0.310	0.310
3	0.260	0.140	0.230
4	0.240	0.190	0.210

Table 1: Silhouette Scores for Different Clustering Methods

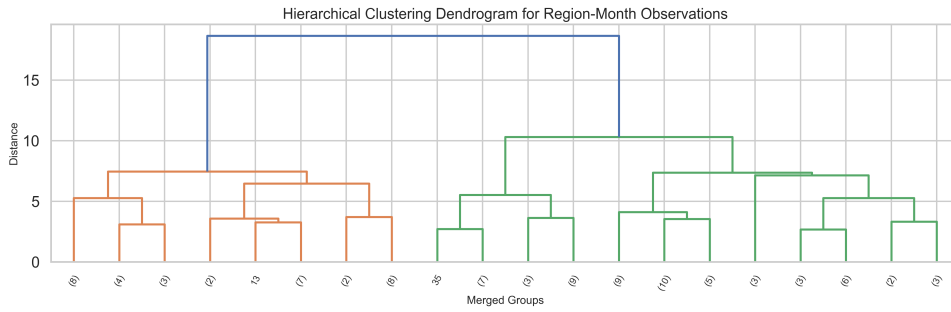


Figure 8: Hierarchical clustering dendrogram

In **figure 8**, the dendrogram shows

- A clear separation is visible at $k = 2$, indicating two main clusters.
- The large vertical gap (distance jump) confirms a strong cluster separation.

So, $k = 2$ is the optimal value for clustering.

Different behavioral patterns exist, but they are not strongly distinct. Both clusters appear in all regions, showing similar behavior patterns globally.

Product Clustering

Now we do product cluster according to total sales and total revenue using the K - means algorithm . From **figure 9** we can see that silhouette score is approximately 0.26, which is moderate but gives stable separation. We can see the clear separation in **figure 10**. We get the best clustering at $k = 2$. From **table 2**, we see that there are two groups,

Cluster 1 (Higher): 47 sales, 10,902 revenue

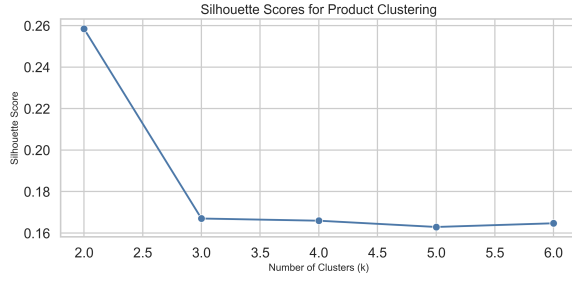


Figure 9: Product clustering diagnostics

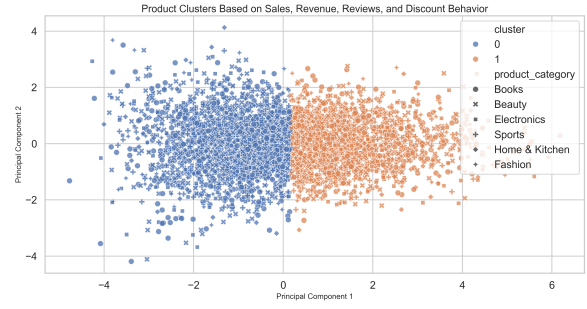


Figure 10: Product cluster visualization

Cluster	Avg Sales	Avg Revenue	Avg Discount
0 (Lower)	29.85	6,147	13.46%
1 (Higher)	47.40	10,902	13.13%

Table 2: Product Cluster using K- means

Cluster 0 (Lower): 30 sales, 6,147 revenue

As sales and revenue are different in both groups, the discount is approximately the same 13%. Products behave differently, but discount patterns remain consistent. Also, we observe that **the difference between these two groups is based on the quantity of sales, not on the quality of product of any category.**

4.3 Time-Series Analysis

Now we have used time Series analysis on a data monthly basis. We see from **figure 11** that the monthly series is fluctuating over time rather than following a consistent trend. And we see that the profit and revenue graphs are the same, this is because profit is 5% of revenue . So, if revenue increases or decreases, the profit graph behaves similarly to the revenue graph.

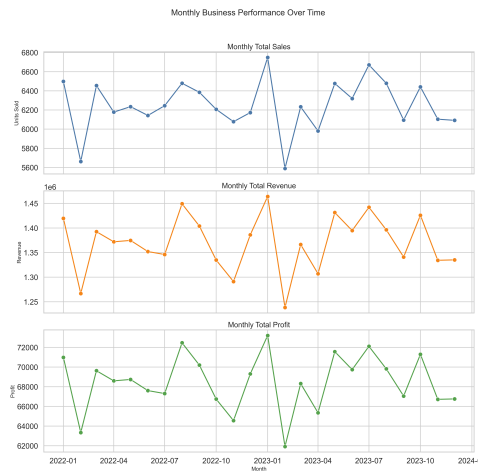


Figure 11: Monthly business performance

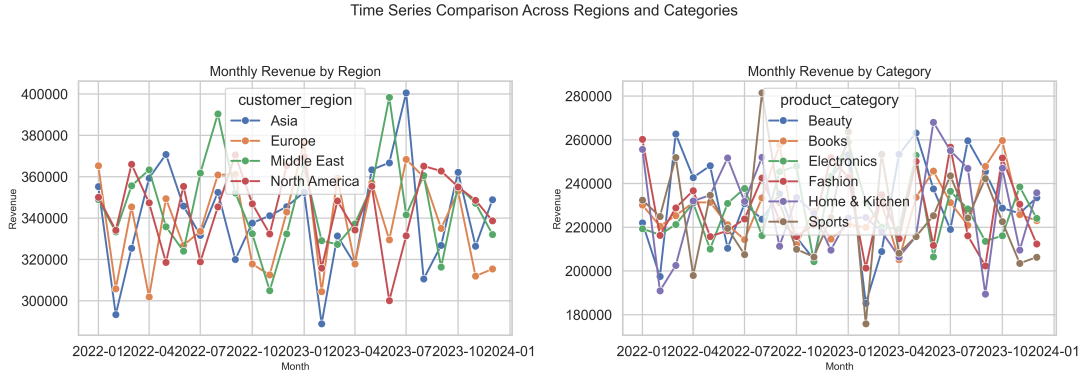


Figure 12: Time-series comparison across regions and categories

we can see from **figure 12**, on a monthly basis the regional revenues mostly stay between 300K–400K and category revenues remain around 200K–260K. We can see that revenue fluctuates over time in all regions and product categories, but there is no clear long-term upward or downward trend. Changes are short-term and are not affected by discounts. Sales remain almost flat, approximately 3 units across all categories and regions. **Revenue consistently decreases with higher discounts in every category; discounts do not increase sales but reduce overall revenue; the impact of discounts is stable over time.**

4.4 Regression Analysis

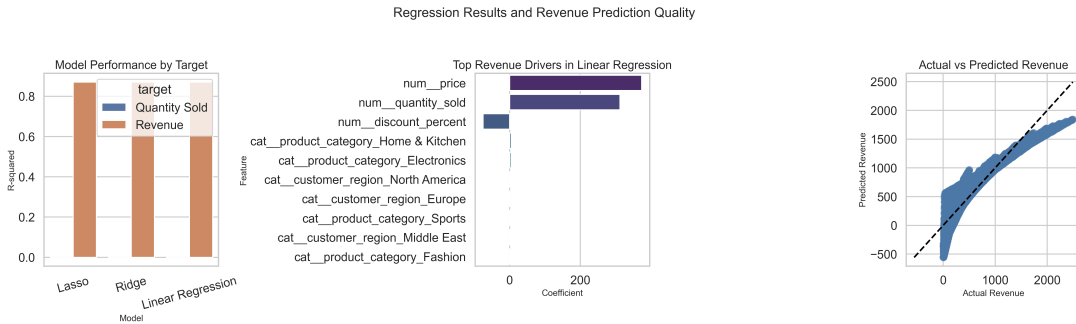


Figure 13: Regression results and model performance

Now we apply different regression models to find the relation between revenue and the other features. we apply linear, ridge, and lasso regression models. We can see from **figure 13** dataset is balanced so the predicted vs actual plot shows a good model fit for revenue. All models give $R^2 = 0.87$ for revenue, which shows that the changes in revenue according to the input variable is 87%. And for sales, all models give $R^2 = 0$, which shows that there is no significant change in sales after discount. Price and quantity sold are the strongest positive factors affecting revenue, whose weight is approximately 300-400 but the discount gives a negative impact to the revenue as its coefficient is -75. **If we increase 1% discount then there is a decrease of 75 units in revenue, a strongly reduced revenue.**

5 Conclusion

In this dataset, discounts decrease total revenue and profit, while sales volume stays almost unchanged; discounting does not generate enough extra demand to compensate for the lower selling price. Overall, discounting reduces financial performance without increasing demand.

According to the investigation, in the real world we suggest that before giving the discount across all the products, we have to monitor how discounts affect revenue and make a strategy to manage profit not only to focus on increasing demand.